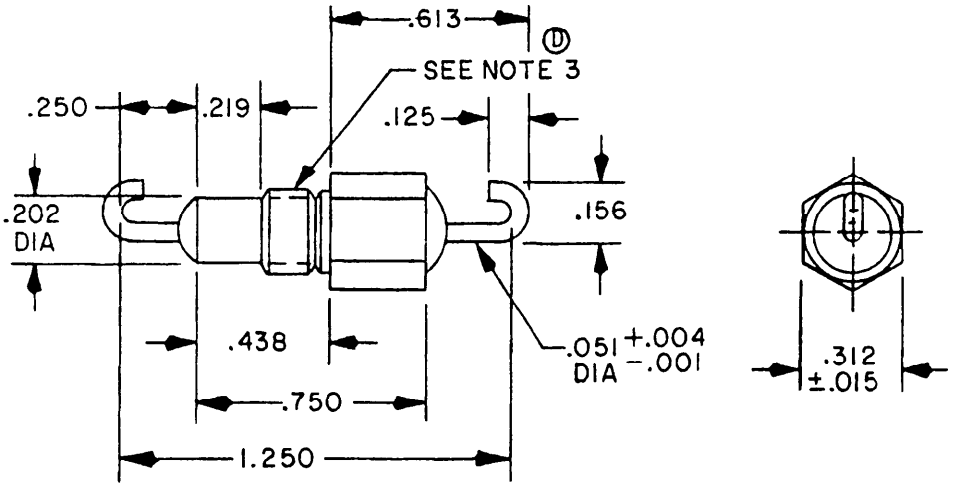
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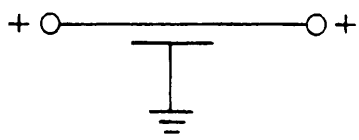
CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE) (FEED-THRU), STYLE CK70

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-11015.



INCHES	MM
.001	0.03
.004	0.10
.015	0.38
.051	1.30
.125	3.18
.156	3.96
.202	5.13
.250	6.35
.312	7.92
.438	11.13
.613	15.57
.750	19.05
1.250	31.75



CIRCUIT DIAGRAM

NOTES:
1. All dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Threaded portion shall be .250-28 UNF-2A and shall extend to within .047 (1.19 mm) of the shoulder.
4. A nut and internal-tooth lockwasher shall be supplied with each capacitor. The nut shall be .312 (7.92 mm) across flats and .125 (3.18 mm) thick. The lockwasher shall be silver plated bronze or brass, .4 (10.2 mm) outside diameter, .26 (6.6 mm) inside diameter.
5. Unless otherwise specified, tolerance is ±.031 (0.79 mm).

Ⓓ FIGURE 1. Dimensions and configurations.

Ⓓ denotes changes

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Feed-thru.

Capacitance value: See table I.

Capacitance tolerance: ±10 percent (K) or ±20 percent (M) as shown in table I.

(D) Rated temperature: -55° to +85°C, voltage derated linearly to 50% at +125°C (see figure 2).

Dielectric withstanding voltage:

Dielectric (DWV): Method 301 of MIL-STD-202.

Test voltage: 2,000 volts dc for 1,000 volt capacitors, or 250 percent of rated voltage for 500-volt capacitors, applied between the case and either terminal.

Body insulation: Not applicable.

TABLE I. Style CK70 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK70AX100-	1,000	AX	10	K, M
CK70AX330-	1,000	AX	33	K, M
CK70AX470-	1,000	AX	47	K, M
CK70AX680-	1,000	AX	68	K, M
CK70AX101-	1,000	AX	100	K, M
CK70AX471-	500	AX	470	K, M
CK70AX681-	500	AX	680	K, M
CK70AW102M	500	AW	1,000	M
CK70AW152M	500	AW	1,500	M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft).

Test potential: 100 percent of rated voltage for 1,000-volt capacitors, or 150 percent of rated voltage for 500-volt capacitors, applied between the case and either terminal.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 20,000 megohms, min, measured between either insulated point and the capacitor case.

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Capacitance (Cap.): Method 305 of MIL-STD-202.

Test frequency:

100 pF or less: 1 MHz ± 1 kHz.

Over 100 pF: 1 kHz ± 100 Hz.

Test voltage: 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 1.5 percent, max (voltage-temp limit X) or 2.0 percent, max (voltage-temp limit W).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

(D) Thermal shock and immersion:

Thermal shock: Method 107 of MIL-STD-202, condition A.

Immersion: Method 104 of MIL-STD-202, condition B.

IR: 3,000 megohms, min.

Terminal strength: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.

DWV: 2,000 volts dc for 1,000-volt capacitors, or 250 percent of rated voltage for 500-volt capacitors.

IR: 3,000 megohms, min.

Cap.: Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

IR: 20,000 megohms, min.

ΔCap.: ±5 percent of initial measurement.

ΔDF: 0.5 percent, max.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).

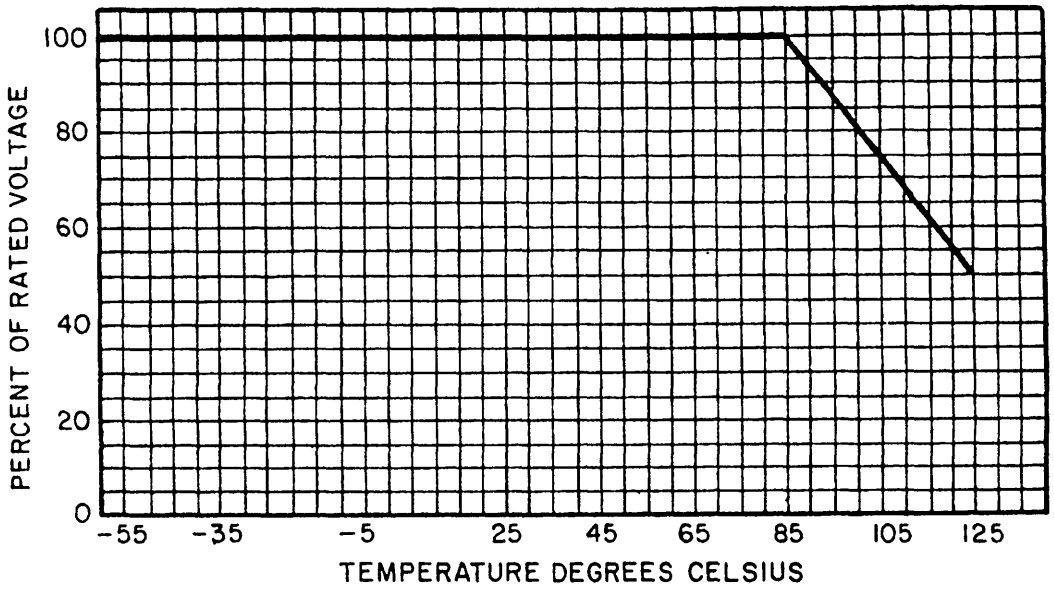
Test potential: 200 percent of rated voltage for 500-volt capacitors, or 150 percent of rated voltage for 1,000-volt capacitors.

IR: 500 megohms, min (at 85°C) and 10,000 megohms, min (at 25°C).

DF: 3.0 percent, max (voltage-temp limit X) or 4.0 percent, max (voltage-temp limit W) (at 25°C).

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Marking: In accordance with MIL-C-11015.



Ⓓ FIGURE 2. Voltage derating with temperature.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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